OM ASHOK DHINOJA

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Work Experience

BrightMinds **Boston, MA**
Manufacturing Engineer **Nov 2025 – Present**

- Reduced average rework by ~30% by adding poka-yoke go/no-go gauges and torque-controlled tools to fix fixture tolerance drift
- Led supplier selection that cut lead time from 8 to 4 weeks, launching a consumer desk robot 30 days ahead of its 180-day NPI plan
- Coordinated a 7-person team and a contract manufacturer through a 100-unit pilot, resolving electromechanical issues via ECOs
- Owned 2 DFM ECOs from design reviews that cut the enclosure from 4 pieces to 3 and moved manuals to a QR code to cut packaging cost
- Managed the desk robot's EBOM and MBOM in Oracle ERP from prototype through pilot and launch
- Ran time studies that removed 3 non-value-added steps, redesigned 2 assembly stations and cut operator wait time
- Built a Power BI KPI dashboard for quality, material flow and project spend, cutting weekly reviews from 30 to 5-7 minutes
- Validated the desk robot's electronics against schematics before the pilot, logging results in controlled test reports

Kanex Fire Solutions **Mumbai, India**
Manufacturing Product Engineer **Aug 2021 – Mar 2022**

- Wrote GMP-compliant SOPs and work instructions that cut new-operator onboarding sessions from half a day to 2 hours
- Designed welding, valve-closing and pressurizing fixtures that cut cycle time from 8 to 6 minutes by fixing tolerance and vibration issues
- Led RCA and corrective action on a welding machine, cutting downtime from 3–4 to 1 hour/week by rewiring it per schematics
- Mapped the value stream across 5 lines and redesigned layouts, raising throughput 30% and cutting transport time 20%
- Saved $0.27 per unit by selecting the motor and designing the mechanism for an automated Powder Filling Machine
- Validated the filling machine against acceptance criteria in a written test plan, producing 85 units per shift vs 35 manually
- Lowered documentation non-conformances from 33% to 12% by maintaining PFMEAs, SOPs and work instructions for 20+ machines
- Executed IQ, OQ and PQ protocols from metal stamping to final assembly under GMP to qualify equipment for production

Projects

UV LED Optical Module and Assembly Fixture

- Cut lens concentricity error from 0.30 to 0.095 mm and seat-depth stack from ±0.20 to ±0.10 mm via GD&T stack-up and an ECO
- Designed a keyed 4-part module, an ESD-safe acetal nest and a go/no-go gauge in SolidWorks so the LED board can't install reversed
- Verified the Rev B design with interference checks before release, catching a 4.43 mm^3 LED-to-lens collision
- Built a 12-mode PFMEA and control plan, cutting projected total RPN from 1,704 to 614 by actioning 9 high-risk modes

Semi-Automated Pothole Filling Machine

- Led a 5-person team to build a pothole-filling machine that filled and leveled test potholes up to 8 in. across
- Engineered a stepper-driven 6-fin rotary dispenser that used ~40% less asphalt than manual filling with 1 operator instead of 3–4
- Coded the IoT system that sized each pothole with VL53L0X sensors and set the dispenser's rotations from its volume

Fire Fighter Robot

- Extinguished 10 of 10 paper fires at 1 to 1.5 m in team testing by building the robot's chassis, driver PCB and wiring
- Designed an L293D/7805 driver PCB in EAGLE that ran both drive motors from an 8–30 V input, then soldered, wired and tested it
- Integrated 3 IR flame sensors, a relay pump and a 50°–130° servo nozzle sweep on an Arduino Nano, tuning a 300 ms approach to the fire

Skills

Manufacturing: Mechanical Assembly, Fixture and Tooling Design, Work Instructions, SOP, Process Flow Improvement, NPI, DFM, DFA, DFMA, DOE, Time Studies, 3D Printing
Quality and Lean: RCA, CAPA, 8D, PFMEA, SPC, ECO, BOM, GD&T, Tolerance Stack-Up, GMP, Poka-Yoke, 6S, Kaizen, Kanban, VSM
Electrical: Schematic Capture and Reading (EAGLE), Electrical Troubleshooting, PCB Design, Soldering and Wiring, Arduino, Sensor and Motor Driver Integration
Software and Tools: SolidWorks, AutoCAD, CATIA, Ansys FEA, ERP, PLM, Power BI, Excel, Word, PowerPoint, Outlook, Minitab, Python
Standards: IS 15683, ISO 9001, ISO 13485

Education

Northeastern University **Boston, MA**
Master of Science, Engineering Management **Sep 2023 – May 2025**
Coursework: Advanced Product Design, Lean Six Sigma, Operations Research, Probability and Statistics

K.J. Somaiya College of Engineering **Mumbai, India**
Bachelor of Technology, Mechanical Engineering **Aug 2019 – May 2023**
Coursework: Additive Manufacturing, Industrial and Engineering Management, Elements of Machine Design, Production Engineering